



Green ENERGY

Like animals, plants need food to **grow** and **survive**. Unlike animals, they can't move around to find it—instead they have to make food themselves. They do this using a process called *photosynthesis*. All they need is **CARBON DIOXIDE**, **WATER**, and **SUNLIGHT**.

LIVING ON SUNSHINE

The Sun is the source of the energy that all organisms need to live. It arrives on Earth as sunlight, although only a tiny fraction is absorbed by plants. They use that energy to power a series of chemical reactions in their leaves that make energy-rich compounds called **sugars**, which can be stored by the plant. When sugars are broken down in the cell they release their energy so that the cell can carry out vital functions. Green algae and some bacteria photosynthesize, too.

Power cells

The reaction that converts light into food happens inside the leaves of plants. The membranes that surround leaf cells are full of tiny structures called chloroplasts. These are where the photosynthesis reaction takes place. Chloroplasts are full of a green pigment called chlorophyll. This is what gives plants their green color.

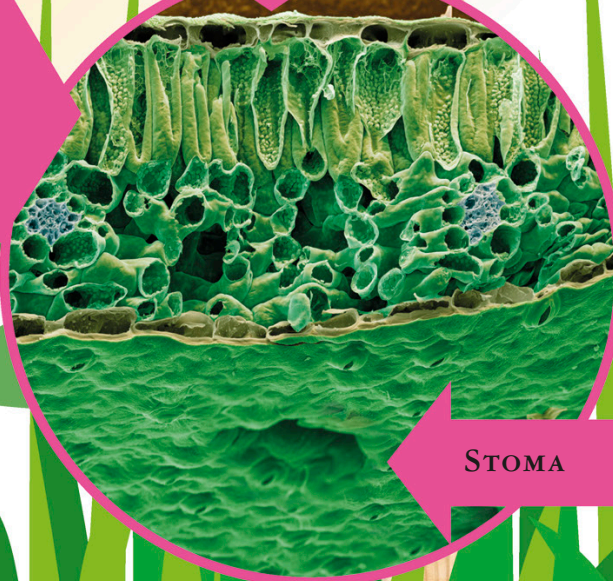


CHLOROPLASTS

Inside the leaf

Air is drawn into the leaf through tiny openings on its underside called stomata (a single one is called a stoma). Air contains only 0.04 percent carbon dioxide, but this is enough for plants to make food. The water needed for photosynthesis to take place is drawn up from the plant's roots and carried through the stem to the leaf cells.

TOP OF
LEAF



STOMA